A Matrix Interpretation of the FFT Algorithm

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Key Concepts

1) The length-N DFT can be described as a matrix operator.

2) The FFT algorithm can be interpreted as factoring the DFT matrix as a product of sparse matrices.

    a) The decimation-in-time FFT algorithm factors the DFT matrix into a permutation matrix that performs the bit reversal stage, followed by $\log_2 N$ matrices that implement the butterflies.

    b) The permutation matrix involves no multiplications as it consists of ones and zeros.

c) The butterfly matrices are sparse and implement N non zero multiplications each.

Video Screenshots

The FFT is a numerically efficient algorithm for computing the DFT

0:14

The DFT can be written as a matrix multiplication

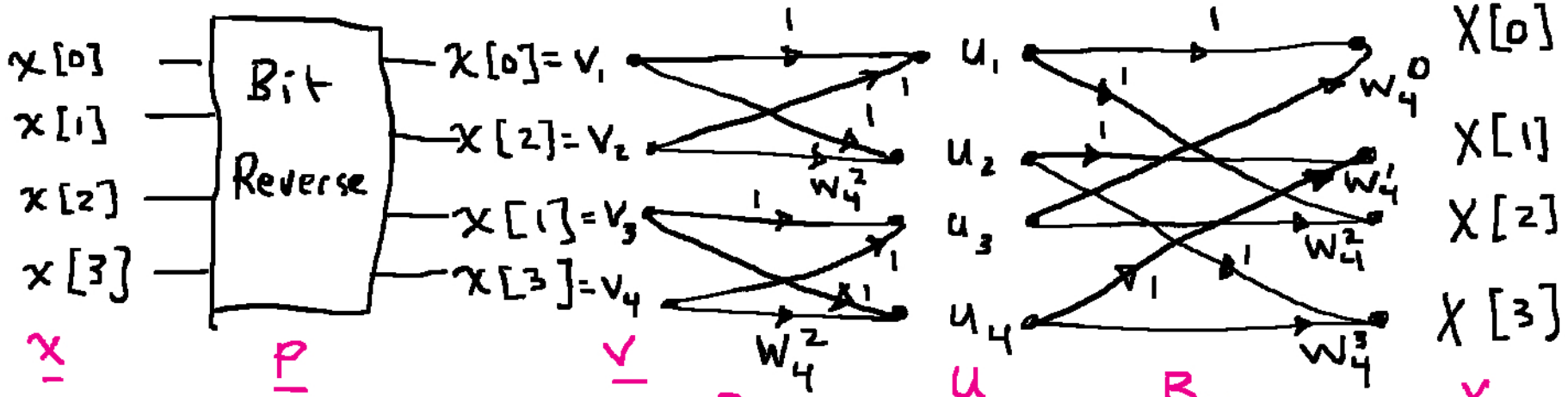
If $\underline{x} = \begin{bmatrix} x[0] \\ x[1] \\ \vdots \\ x[N-1] \end{bmatrix}$, $\underline{X} = \begin{bmatrix} X[0] \\ X[1] \\ \vdots \\ X[N-1] \end{bmatrix}$, then $\underline{X} = \underline{W}\,\underline{x}$

$$\underline{W} = \begin{bmatrix} W_N^0 & W_N^0 & \cdots & W_N^0 \\ W_N^0 & W_N^1 & \cdots & W_N^{N-1} \\ \vdots & & & \\ W_N^0 & W_N^{N-1} & \cdots & W_N^{(N-1)^2} \end{bmatrix}, \quad W_N = e^{j\frac{2\pi}{N}}$$

The FFT algorithm factors $\underline{W}$ into a product of simple matrices

FFT Algorithm - Factor the DFT Matrix $\underline{W}$

Ex. $N = 4$



$\underline{x}$ $\underline{P}$ $\underline{v}$ $\underline{B}_1$ $\underline{u}$ $\underline{B}_2$ $\underline{X}$

$\underline{X} = \underline{B}_2 \underline{u}$ $\quad$ $\underline{u} = \underline{B}_1 \underline{v}$ $\quad$ $\underline{v} = \underline{P}\underline{x}$

$$\underline{B}_2 = \begin{bmatrix} 1 & 0 & W_4^0 & 0 \\ 0 & 1 & 0 & W_4^1 \\ 1 & 0 & W_4^2 & 0 \\ 0 & 1 & 0 & W_4^3 \end{bmatrix} \qquad \underline{B}_1 = \begin{bmatrix} 1 & W_4^0 & 0 & 0 \\ 1 & W_4^2 & 0 & 0 \\ 0 & 0 & 1 & W_4^0 \\ 0 & 0 & 1 & W_4^2 \end{bmatrix} \qquad \underline{P} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$\underline{X} = \underline{W}\underline{x} = \underline{B}_2 \underline{B}_1 \underline{P}\,\underline{x} \qquad \underline{W} = \underline{B}_2 \underline{B}_1 \underline{P}$$

$N \log_2 N$ mults

In general, $\underline{W} = \left(\prod_{k=1}^{\log_2 N} \underline{B}_k \right) \underline{P}$

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